Post-Scarcity Simulation V2

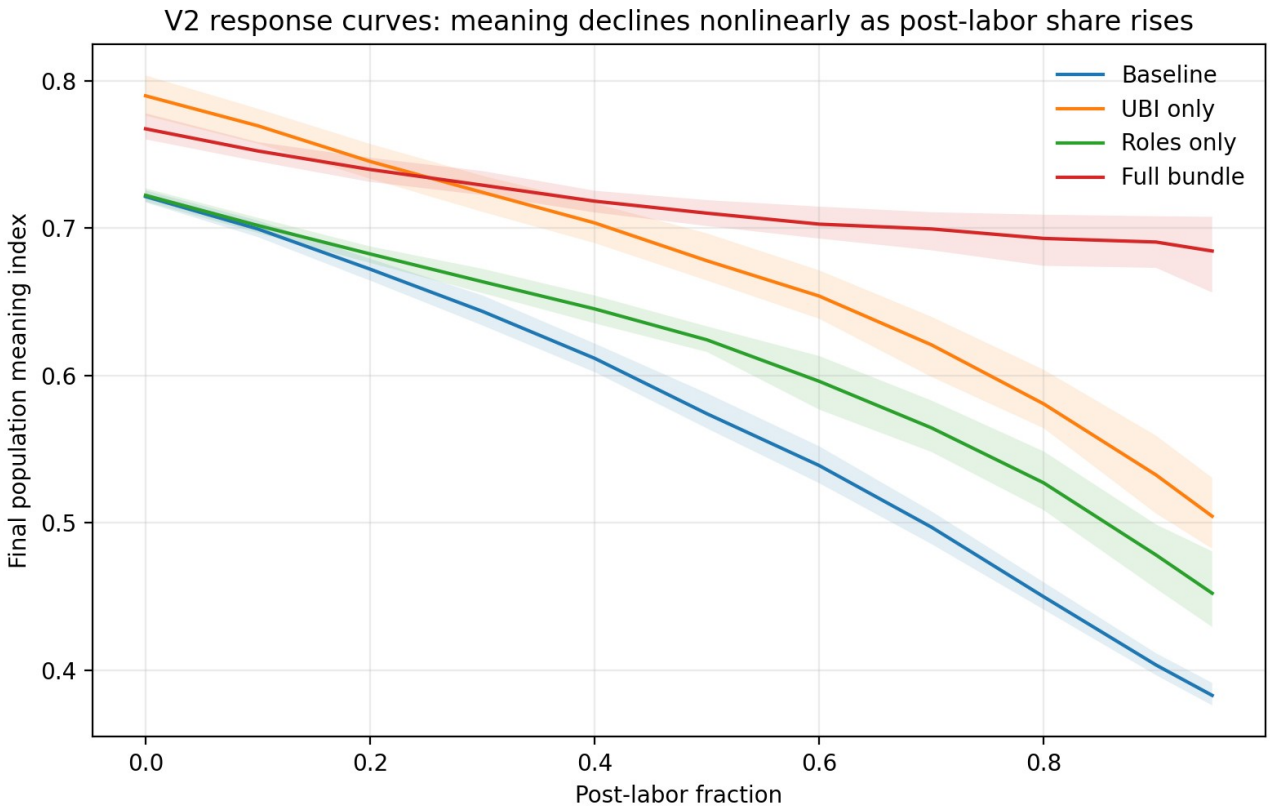
Mechanism-transparent agent-based model with 100-run Monte Carlo, contagion sensitivity, intervention decomposition, and network topology checks

Generated from the V2 run on 2026-03-13

Executive summary

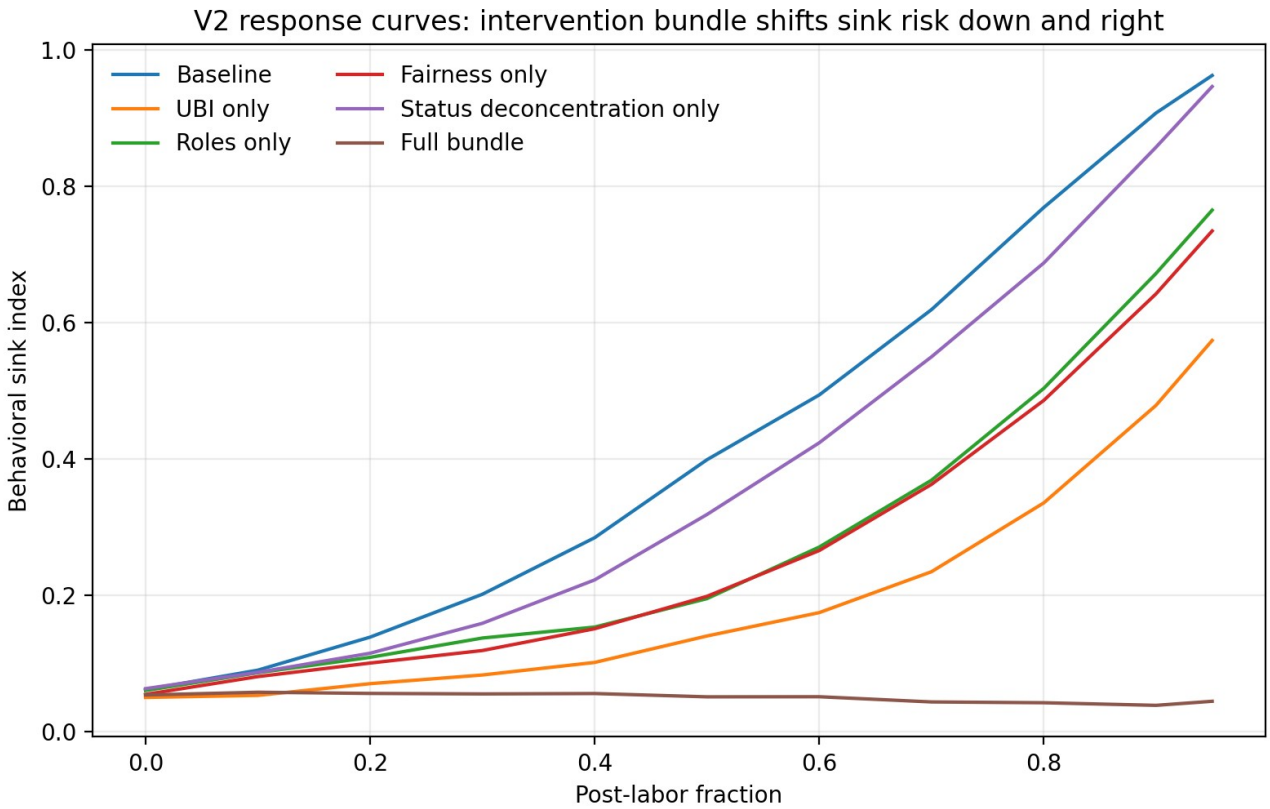
- The V2 model still shows nonlinear deterioration under post-labor conditions, but the result is now more transparent: explicit psychological state updates, endogenous role search, heterogeneous agent types, and separate contagion/network tests all preserve the core pattern.
- In the baseline small-world model, the population meaning index falls from 0.672 at 20% post-labor to 0.574 at 50%, then drops to 0.450 at 80% and 0.383 at 95%. The behavioral sink index climbs from 0.139 to 0.399, then accelerates to 0.769 and 0.963.
- At 80% post-labor, the baseline scenario enters a collapse region: 97% of runs finish with sink index above 0.7. The full intervention bundle keeps the same system far from collapse, with meaning 0.693, sink 0.042, and zero collapse runs in this sample.
- Decomposing the intervention bundle shows that redistribution/fairness and income support matter at least as much as role substitution in this parameterization; status deconcentration alone helps least. So V2 does not support a simple 'roles only' story.
- Contagion strength changes severity materially, especially near 80% post-labor, but the baseline threshold does not disappear when contagion is weakened. Network topology matters less than expected in this version; the displacement level is doing most of the work.

Figure 1. Meaning response curves



Meaning declines smoothly at low displacement, then more sharply in the baseline between roughly 0.7 and 0.8 post-labor. UBI-only and the full bundle push the curve upward, while the full bundle nearly flattens it across the tested range.

Figure 2. Behavioral sink response curves



The sink curve is the more policy-relevant counterpart to the meaning curve. The baseline crosses into high-risk territory around 0.8. The full bundle shifts the entire sink curve downward; UBI-only, fairness-only, and roles-only all help, but to different degrees.

Model changes from V1 to V2

- Explicit mechanism: autonomy, competence, and relatedness are now primary psychological variables. Status recognition and role access are modeled as mediators. Meaning is an outcome variable, not a free-floating score.
- Heterogeneity: agents are sampled into resilient, balanced, and vulnerable profiles with different resilience, social capital, exploration, and skill-transferability parameters.
- Adaptation: non-producer agents can search for and accumulate substitute-role access endogenously; role access is no longer purely handed down by policy.
- Sensitivity and robustness: the model adds a separate contagion sweep, network topology checks (ring, small-world, Barabási-Albert), and 100 Monte Carlo runs per core parameter point.

Core update structure

- Autonomy_t+1 = f(base autonomy, role access, redistribution fairness, resilience, local engaged / withdrawn / collapsed exposure, status gap)
- Competence_t+1 = f(base competence, necessary production role, role access, transferability, local exposure)
- Relatedness_t+1 = f(base relatedness, social capital, role access, redistribution fairness, local aggression / collapse)
- Status_t+1 = f(necessary role, role access, redistribution fairness, relatedness, status concentration, local aggression)
- Meaning_t+1 = f(autonomy, competence, relatedness, status, role access, role loss, status gap, contagion exposure, resilience, redistribution fairness)

Simulation settings

Element	Value
Population	180 agents per run
Time horizon	70 steps
Main curve sweep	11 post-labor fractions from 0.00 to 0.95
Monte Carlo	100 runs per main parameter point
Core scenarios	Baseline, UBI only, Roles only, Fairness only, Status deconcentration only, Full bundle
Sensitivity studies	Contagion low/medium/high at 0.5 and 0.8; network ring/small-world/Barabási at 0.5 and 0.8
Total runs executed	9,120 runs including time-series exemplars

Key point table

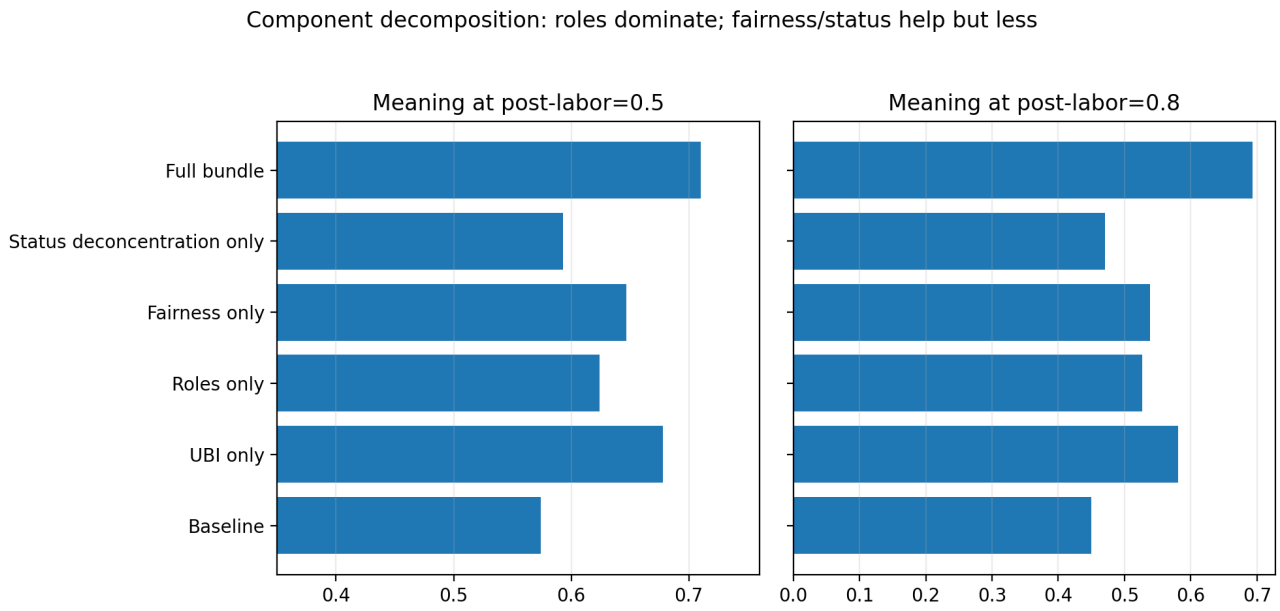
Scenario	Post-labor	Meaning	Sink	Collapse probability
Baseline	0.20	0.672	0.139	0.00%
Baseline	0.50	0.574	0.399	0.00%
Baseline	0.80	0.450	0.769	97.00%
Baseline	0.95	0.383	0.963	100.00%
UBI only	0.80	0.581	0.336	0.00%
Full bundle	0.80	0.693	0.042	0.00%
Full bundle	0.95	0.685	0.044	0.00%

Findings

1. Baseline nonlinear deterioration

Baseline deterioration is not linear. The steepest decline in the meaning curve occurs between 0.70 and 0.80 post-labor. At 0.50, the system is degraded but not collapsed (meaning 0.574, sink 0.399). At 0.80, the same system crosses into a high-risk region (meaning 0.450, sink 0.769), and 97% of runs end above the sink-collapse threshold.

2. Intervention decomposition

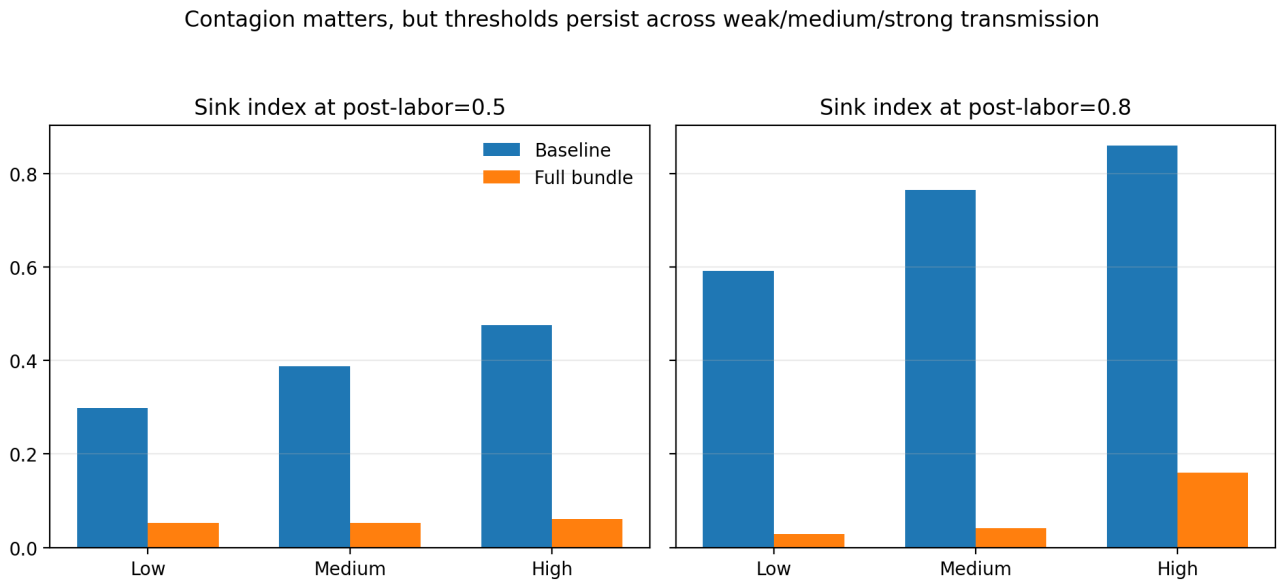


The decomposition result is not the same as V1. In this V2 parameterization, UBI/fairness-type support does more work than expected, and status deconcentration alone does the least. At 80% post-labor, the meaning gains relative to baseline are approximately +0.131 for UBI-only, +0.088 for fairness-only, +0.077 for roles-only, and only +0.020 for status deconcentration alone. The full bundle is strongest because it changes all three channels together.

3. Behavioral composition at 80% post-labor

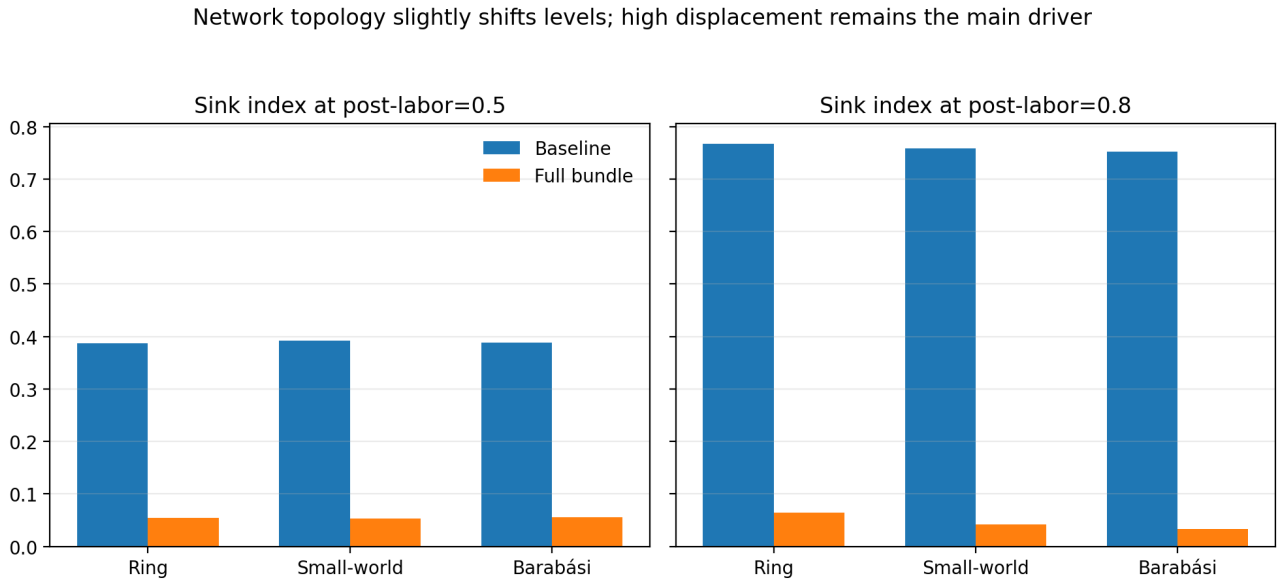
Scenario	Engaged	Drifting	Withdrawn	Aggressive	Collapsed
Baseline	0.000	0.248	0.708	0.040	0.005
UBI only	0.067	0.608	0.294	0.031	0.000
Full bundle	0.416	0.545	0.027	0.011	0.000

4. Contagion sensitivity



Contagion clearly matters. In the baseline at 80% post-labor, the sink index is about 0.593 under low contagion, 0.766 under medium contagion, and 0.861 under high contagion. That said, the threshold is not a pure contagion artifact: weakening contagion reduces severity, but the baseline still degrades sharply near 80% post-labor. Under the full bundle, the same contagion sweep leaves the system much more stable.

5. Network topology checks



Topology shifts levels modestly rather than overturning the story. At 80% post-labor in the baseline, sink outcomes are similar across ring (0.767), small-world (0.758), and Barabási-Albert (0.752) networks. In other words, V2 is still primarily displacement-driven. This could change once stronger hubs, modular community structure, or targeted institutional nodes are added.

Interpretation, caveats, and next moves

What V2 now supports

- The baseline threshold is not just an artifact of a ring network or a tiny Monte Carlo sample; it persists under 100-run sweeps, contagion changes, and alternative topologies.
- The critical region in this model is around 70–80% post-labor for the baseline configuration.
- Intervention effects are decomposable. In this parameterization, redistribution/fairness support is at least as important as role creation, and status deconcentration alone is weak.
- The model no longer assumes passive decline only: endogenous role search and agent heterogeneity are now in the loop.

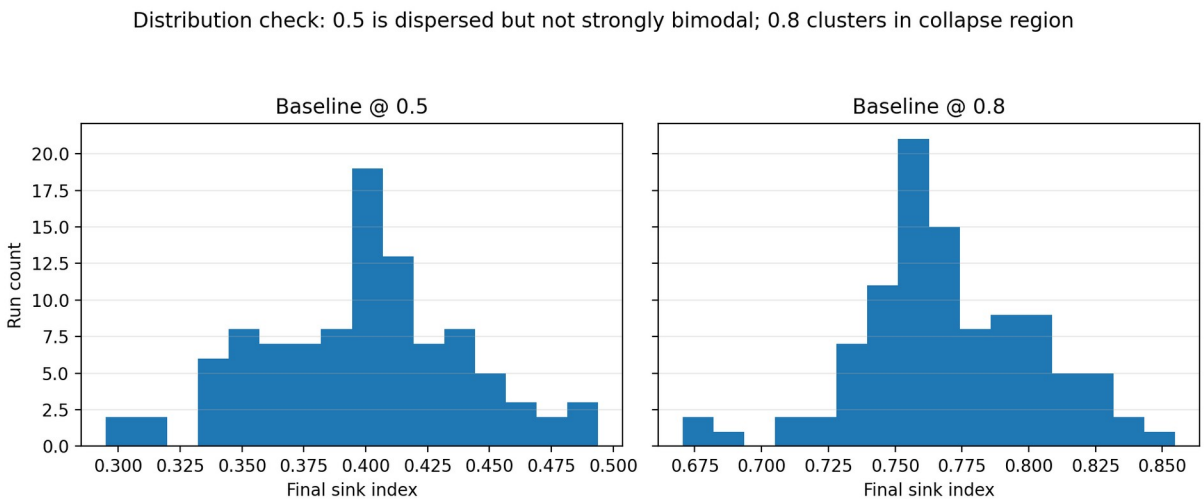
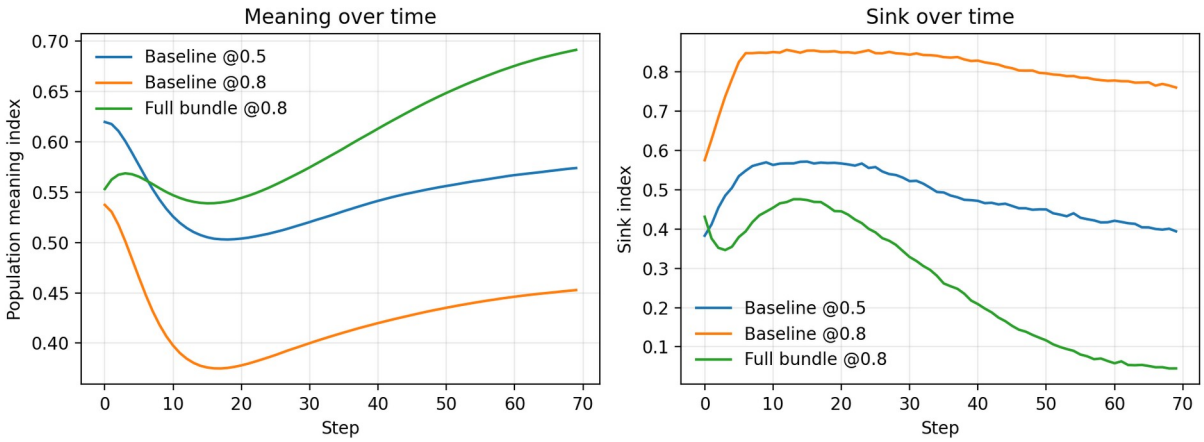
What V2 still does not support

- This is still a computational thought experiment, not an empirical forecast about real societies.
- The full bundle is probably too strong in this parameterization; its near-flat curve at 95% post-labor is a sign that those policy coefficients should be stress-tested further.
- The network layer is still relatively simple: fixed degree, no endogenous rewiring, no formal communities, and no institutional super-nodes.
- Birth intention is modeled as a vitality indicator, not yet a full demographic feedback system.

Recommended V3 priorities

1. Turn role access into multiple explicit channels: civic, care, art/creation, and co-ownership.
2. Add community modules and institutional hubs so topology can matter more realistically.
3. Run a stronger global sensitivity design over the policy coefficients, especially because UBI/fairness came out stronger than expected in V2.
4. Close the demographic loop so persistent low meaning and low birth confidence feed back into long-run social structure.

Appendix figures



Time paths: the bundle slows and partially stabilizes degradation at high displacement